

Coarse sediment connectivity in river channel systems: a conceptual framework and methodology

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Received 26 February 2002; received in revised form 1 August 2002; accepted 3 January 2003

Abstract

Identification of coarse sediment connectivity down river-channel systems is important for understanding the linkages between river reaches, the influence of sediment sources on channel morphology and the mechanisms and liability to propagation of morphological change. A conceptual framework is developed in which several types and degrees of connectivity are proposed. It is hypothesised that sediment connectivity varies with sediment sources and with stream competence to transport the coarse fractions. A method of identifying the status of reaches is developed, based on field mapping of morphology and sedimentology. The analysis is exemplified by two channels: one an ephemeral channel in SE Spain; the other a perennial channel in NW England. Questions arise over whether reaches that lack channel stores transfer coarse sediment. Three scenarios are suggested: (1) lack of flux due to lack of competence of the reach; (2) flushing through of sediment due to high competence; and (3) potential transport of coarse sediment but exhaustion or lack of availability. Changes in connectivity will take place over different time scales depending on the nature of exchange and time scales of storage. Short-term variations in connectivity can be identified by repeated mapping.

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Keywords: Fluvial morphology; Bed material; Sediment transport; Connectivity; Channel bars

1. Introduction

Ascertaining the degree of sediment connectivity within a river system is vital for understanding how the system evolves and functions and for interpreting the morphology and dynamics of the channel. Identification of connectivity focuses attention on the spatial relations of sediment transfer within a river channel. It is also important in channel management

for understanding the impacts of direct and indirect action on the channel and for habitat maintenance.

Connectivity in this context means the physical linkage of sediment through the channel system, which is the transfer of sediment from one zone or location to another and the potential for a specific particle to move through the system. This entails understanding the local sediment sources and the mechanisms, conditions, routes and distances of transport. It means identifying the sediment storage zones. Ideally, the duration of storage of specific particles needs to be known to distinguish short-term stores from longer-term 'sinks'. Connectivity can be set in

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the context of a sediment budget, with sediment inputs and outputs within and between reaches. Channel morphology and stability could be expected to reflect the net sediment budget, with evidence of net erosion, net aggradation or approximate balance. The emphasis here is on the physical connection and transfer of sediment. In this, it is comparable with the use of the term connectivity in ecology and in hydrological applications to river systems (e.g. Ward and Stanford, 1995) and is coming into use in geomorphology (e.g. Fryirs and Brierley, 2000; Harvey, 2000) at scales ranging from reaches to basins.

A general assumption is that the channel functions as a system and that sediment is moved through that system. A high degree of connectivity is generally assumed. Whilst this may hold true for fine sediment (sand and finer), there is increasing evidence that both sources and budgets of coarse sediment may be much more localised. For example, on channels with very active reaches, containing many channel bars, that are succeeded downstream by apparently inactive reaches, with few signs of sediment storage, the juxtaposition of these contrasting reaches raises questions of the extent to which coarse sediment is carried through the apparently inactive reach and the extent to which upstream reaches are affecting those downstream. That coarse sediment budgets may be localised is also evident in the often close association between sources of sediment and zones of deposition. It should, however, also be recognised that fluvial systems are dynamic and that the degree of connectivity can change over time. Coarse sediment is important because of its influence on channel morphology (Leopold, 1992; Friend, 1993), although it may comprise only a small portion of total sediment load.

Until recently, much of the emphasis within the coarse sediment literature was on sediment fluxes, their measurement and modelling, mainly from the viewpoint of the mechanics of sediment transport. Over the last decade, a much more overtly geomorphological approach has been apparent with the interaction of sediment dynamics and channel morphology becoming integral to studies (e.g. Macklin and Lewin, 1989; Goff and Ashmore, 1994; Ashmore and Church, 1998). The morphological approach to the estimation of sediment fluxes has produced very fruitful results (e.g. Lane et al., 1995; Martin and Church, 1995; Ham and Church, 2000). Although

similar qualitative approaches were suggested long ago (Popov, 1962; Neill, 1971), they “have not been widely pursued” (McLean and Church, 1999) until recently when technological developments that allow rapid detailed surveys and construction of DEMs have facilitated quantitative studies. Much more work has also been done on the detailed kinematics of coarse sediment, particularly on sediment routing (e.g. Nicholas et al., 1995), and distance of travel (e.g. Hassan et al., 1991). Investigations of the links between bars and differential transport and deposition (Wathen et al., 1997; Wathen and Hoey, 1998) and residence times (Kelsey et al., 1987) have resulted in a much greater understanding of within-reach behaviour.

The coupling of sediment systems has become a focus of research, in particular the coupling between hillslopes and channels. Brunsden (1993) recognised three types of coupling and Harvey (1997, 2001) has demonstrated the dynamic nature of coupling and its importance in terms of buffering and sensitivity of systems. Harvey (2001) has discussed and exemplified reach-to-reach coupling, i.e. channel connectivity. The occurrence of reaches with large volumes of sediment stored in bars was recognised by Church and Jones (1982) who termed them ‘sedimentation zones’. They noted that they are zones of channel instability and that the zones may be mobile, representing a large wave of sediment moving through a system. The overall behaviour of sediment has also been investigated in terms of sediment waves (sediment slugs) moving through systems by, for example, Nicholas et al. (1995) and Madej and Ozaki (1996). Within-reach behaviour and evidence of fining and selective transport (e.g. Sambrook Smith and Ferguson, 1995; Ferguson et al., 1998) have received much attention. Consideration of the mechanisms within reaches has been combined with the concept of sediment links (Rice and Church, 1998; Rice, 1998) to elucidate the role of tributaries in influencing down-channel sediment characteristics. Goff and Ashmore (1994), Madej and Ozaki (1996), and Martin and Church (1995) amongst others, have demonstrated the differential functioning of within-channel zones or reaches. The longer-term dynamics and functioning of the whole system have been elucidated, for example, in a series of papers on a disturbed catchment in New South Wales by Fryirs and Brierley (1999, 2001).

2. Conceptual framework and hypothetical models

The degree of connectivity may vary between systems. Here, a conceptual framework for different types and degrees of coarse sediment connectivity in river channels is put forward. In particular, the hypothesis that some reaches may be poorly connected for coarse sediment transport and that behaviour and budgets may be localised for those reaches is examined.

In this paper, a distinction is made between sediment connectivity and sediment continuity. *Connectivity* is when particles physically pass through the channel system (in which case source materials from the upper part of catchment should be identifiable). *Continuity* is when a particular size-fraction is present throughout the channel but the particles are actually exchanged, with deposition on bars and fresh material of similar nature being eroded from the banks. Sediment connectivity in this context means that a particle has the potential to be transported all the way through a well-connected channel system. Sediment continuity means that there must be a balance between inputs and outputs and material entering and leaving storage zones. A budget would indicate the amounts in each of those components. Generally, continuity and budgets are calculated for the whole sediment load rather than various size fractions. Although continuity means that there are the equivalent amounts in the various components, it does not necessarily denote that the sediment has physically come from a particular source or zone.

It is generally assumed that, in the long term, coarse material can be transported through most channel reaches. When it cannot be transported through a reach, the material is either broken down in situ until it can be transported or the channel morphology adjusts until the reach is competent. However, there are channel systems in which it is doubtful whether coarse material ever gets through particular reaches, for example reaches that lack bars and are much longer than reported step lengths of transport. Therefore, a major question is whether there are reaches through which, with the present channel morphology, sediment supply and discharge regime, the coarsest fractions of the sediment load are not carried. If so, and the upstream reach is carrying such material, then it might be expected that accumulation

would take place at the head of the nontransporting reach, forming a ‘sedimentation zone’ (Church and Jones, 1982). Lack of net sedimentation implies either no coarse transport into that reach or that the erosion and deposition are balancing out upstream. It is recognised that coarse sediment will move only in times of flood or competent high flows. The particles will move in steps with periods of storage of varying lengths of time between movement. Connectivity will vary with the size of material and the discharge regime, the coarsest material moving less frequently and, generally, shorter distances than the finer material.

Analysing geomorphic systems in terms of sediment sources, transfers and sinks has long been applied to geomorphic systems but has more recently been developed into consideration of sediment budgets by quantifying components, for example by Reid and Dunne (1996) in fluvial systems, and Bray et al. (1995) and Komar (1996) on the coast. Various types of sources, stores and sinks have been recognised in the literature. The focus here is on *channel* sediment connectivity and therefore sources have been classified into within and from outside the channel, rather than into hillslope and channel or alluvial and non-alluvial (Table 1). Tributaries are difficult to deal with because they can constitute the channel system farther

Table 1
Example coarse sediment sources, stores and sinks within a channel reach of a typical river system

Sources		Stores		Outputs
External	Internal	Short term	Long term	
Tributary streams	Sediment from channels	Bars	Floodplain	Sediment flux downstream
Alluvial bank erosion	upstream Bar erosion Bed erosion	Bed		
Valley wall/colluvial erosion				
Landslides/mass movements				
Hillslope erosion by overland flow, piping and gullyng				

upstream. However, they are treated here as an external input in relation to any particular channel reach. The alluvial sources do not add new material to the system in the long-term since, by definition, the material must have already been transported by fluvial processes. The connectivity is, however, considered for each channel reach. Therefore, any source of material outside that reach of the channel is an external source. Sediment within the channel, either on the bed or on bars is that which can provide the connectivity under the present morphology. With channel migration or changes in the channel morphology, other longer-term alluvial stores can potentially be tapped by bank erosion of the floodplain or of alluvial terraces. The difference between stores and sinks is a matter of time scale and therefore the categories overlap and the status depends on the period of storage in a particular system and on the relevant time scale of analysis. In terms of stores and sinks, again several have been previously recognised in fluvial systems but the classification used here is presented in Table 1.

The degree of connectivity in a system depends on the routing of sediment between sources, stores and sinks, and output. If sediment is moving from either an external or internal source to a store downstream and that store acts as an internal source for the downstream reach, then the system is connected. If,

however, the sediment is deposited on a bar or the floodplain and goes into sufficiently long-term storage that it becomes a sink, then the system is poorly connected. In the short-term, the same particles may not be transferred completely through the system; if sediment is carried directly from a source to the outlet of the system, as is assumed happens for much suspended sediment in a flood event, then the system is fully connected for that sediment. The connectivity is therefore defined for a particular size fraction.

In the short-term, it is the channel bars and the bed that provide coarse sediment connectivity and thus their characteristics are of key importance. Local input to the reach from the banks, whether by erosion of alluvial or nonalluvial material, is an external source. Input from upstream of a reach is a channel source to that reach. Therefore, identifying the connectivity is a matter of identifying the extent that coarse sediment is moving from bar to bar, or bar to bed, or bed to bed, or bed to bar. The crucial element is the extent to which a bar or bed is being eroded. Hypothetical models of different relations are shown in Fig. 1. Transfer and connectivity will cease if the sediment goes into long-term storage on the bar, usually entailing lateral channel movement or net aggradation or narrowing. Connectivity is established when there is renewed transport from that store, which might be after tens of years in a very mobile channel but

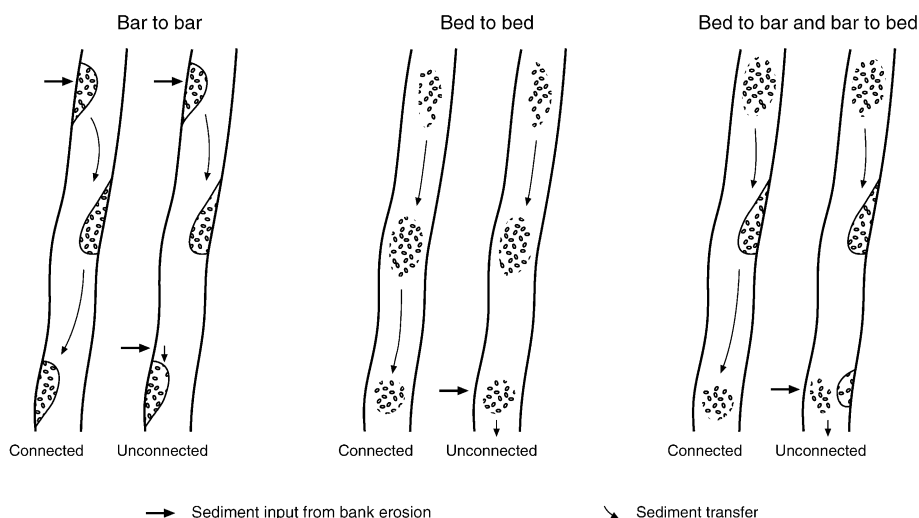


Fig. 1. Hypothetical coarse sediment linkages within the channel.

thousands of years in laterally stable channels (Fig. 2). Thus, connectivity varies with the time scale of analysis.

Crucial to these inferences is the idea that the presence of bars of coarse sediment in a reach is indicative that coarse sediment transport is taking place. Conversely, if there is an absence of bars in a channel or of sediment on the channel bed, the implication would be of a lack of coarse sediment transport. However, a lack of coarse sediment stores within a reach may not always be indicative of a lack of sediment transport. It is hypothesised that there are three conditions in which lack of coarse sediment stores could occur (Fig. 3):

1. The reach is of low competence and coarse material cannot be transported through the reach.
2. The reach is highly competent and coarse material is flushed through the reach in high flows but because of its high power, the material is not deposited within the reach.
3. The reach would be competent to transport coarse material but such material is not available due to lack of supply.

Each of these situations could be expected to have associated indicators and features that would provide clues to the likely condition (Fig. 3). The emphasis is on the transport and deposition of the coarsest fraction of the load. Finer sediment may still be deposited. Thus, the bed in such reaches would be expected to be

composed of finer material than other reaches or to be free of sediment.

In a reach with bars and active bank erosion, then the budgets may still be localised and sediment may not necessarily be transferred from bar to bar, only from bank to bar. If a series of coarse bars is present but no input sources of sediment are present, then there must be connectivity from bar to bar to supply the bars downstream. Scour of the bed could supply limited material but would soon become exhausted and lead to lack of connectivity. Net deposition into a zone means there would have to be morphological adjustment either by net aggradation or by lateral channel movement through erosion, in which case there would be a new sediment source. However, in some cases, sediment continuity and energy relations could be satisfied by exchange of sediment sizes, the coarse material being deposited and finer material supplied to downstream reaches (Fig. 4).

Thus, a series of hypothetical models can be proposed representing different degrees of connectivity. These can be considered in terms of spatial relations under existing morphology, sediment supply and discharge regime. They can then be considered temporally, in how connectivity changes with alterations of each of those controls. The connectivity of the system will vary over time as supply varies and as the episodicity or frequency of transport of certain fractions varies. Repeated surveys of channels would allow changes in connectivity to be monitored. Long-term variations could be hypothesised in relation to

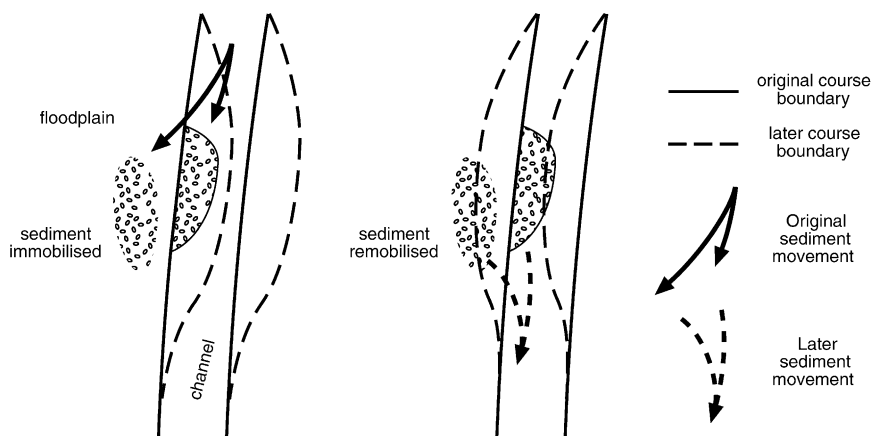


Fig. 2. Conditions for immobilisation and remobilisation (connectivity) of sediment associated with longer-term storage and channel movement.

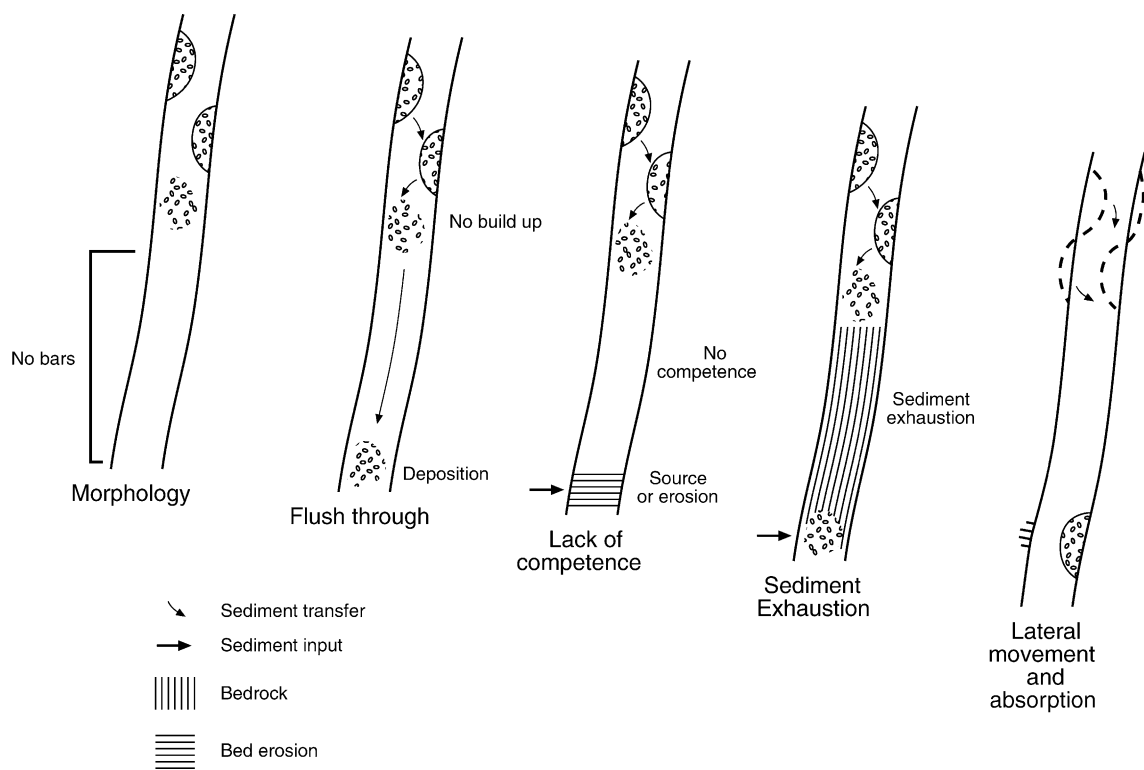


Fig. 3. The range of possible conditions for reaches that lack coarse sediment stores.

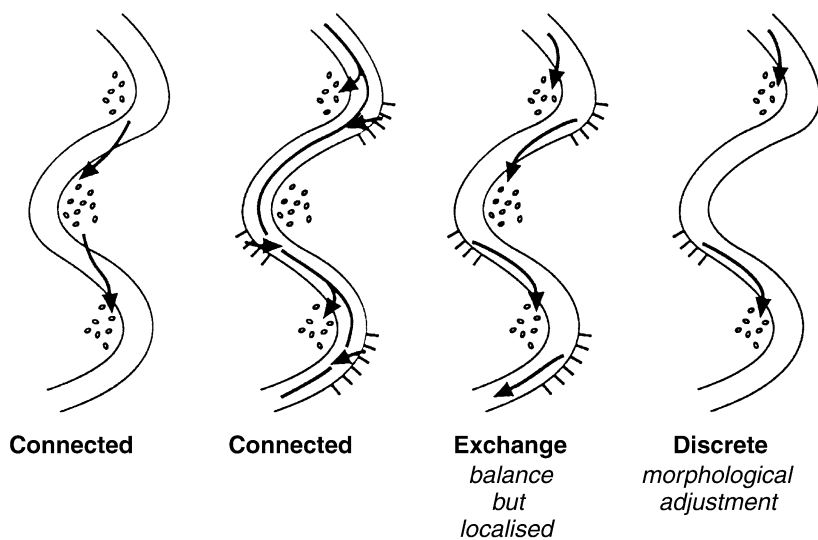


Fig. 4. Possible sediment transfer relations in a meandering channel.

periodic or sudden inputs of sediment or inputs under different conditions, e.g. from previously active hill-slope sources. Likewise, transfer under a different discharge regime could be hypothesised and the implications for connectivity examined. In a continuous alluvial system, excluding non-alluvial external sources, four different time scales could be applicable or distinguishable: (i) single contemporary state; (ii) connectivity over a series of events not involving lateral channel movement; (iii) time scale of erosion of previously deposited floodplain, which could be of various time periods depending on the age of floodplain being reworked and the dynamism of the channel; (iv) time scale of erosion and reworking of alluvial terrace deposits.

Five different types of coarse sediment connectivity can be identified (Table 2). In application of this classification, both sediment size and time scale of analysis need to be made explicit. The classification has been designed for initial application to on-off contemporary surveys and assessments of the state of connectivity in a channel.

The five classes are as follows:

Unconnected systems—localised sources and budgets in which reaches are operating almost independently of each other. Propagation of effects from one reach to another takes a long time or requires a very large event or some external impetus such as interference with the channel. Lack of connectivity may be due to a lack of competence to transport coarse sediment through the linking reaches. Coarse sediment transport will occur in reaches downstream only if there is a new source of coarse sediment.

Partially (or episodically) connected systems—systems in which there is little coarse sediment transport between reaches except in extreme events. The size of events and the size fraction involved need to be

made explicit in any application. All channels are likely to show some partial connectivity since fractions of the load will be transported only by large flood events. This category is distinguished from unconnected systems in that there is evidence that the coarsest material is occasionally transported.

Potentially connected systems—systems in which coarse sediment could move from one reach to another but does not actually do so because of lack of supply of coarse sediment. The connectivity could easily change with fresh inputs of sediment from upstream or as stores fill up and a progressive wave moves down through the reach.

Connected systems—systems in which coarse sediment moves easily and frequently through the system, transported by ‘normal’ flood events, i.e. with recurrence intervals of ca. 0.5–5 years. Sediment may be temporarily stored on bars but is easily remobilised. Variations in channel morphology and to some extent supply mean that actual amounts and sizes of coarse sediment transport vary.

Disconnected system—system in which barriers to movement have been constructed, e.g. dams and weirs, preventing downstream movement of coarse sediment within the lifetime of the structure.

In order to recognise these different types or degrees of connectivity, the spatial relations of the sources and stores and the field evidence for processes and transfers must be recorded and analysed.

3. Methodology

The main principle of the proposed method is that the morphology and sedimentology are the net outcome of the processes operating over a period of time and thus processes can be inferred from consideration of morphology and sedimentology. The morphology and sediment calibre are recorded by field mapping. The spatial relations between reaches can then be assessed.

The aims of the mapping are to identify inputs of coarse sediment to the channel system, zones of storage, signs of sediment transfer and the net functioning of the channel reach. Morphological mapping requires geomorphological expertise; however, recognition of erosional and depositional features, their relations and relative age are techniques that are familiar and standard to geomorphologists. It is

Table 2
Classification of channel reach connectivity

Unconnected	Local sources and stores/sinks. Incompetent reaches between
Partially connected	Transfer only in extreme flood events
Connected	Coarse sediment transfer during ‘normal’ flood events
Potentially connected	Competence to transport but lack of supply
Disconnected	Formerly connected but transfer is now obstructed (e.g. by dams)

assumed that certain features can be recognised that indicate erosional or depositional activity.

Most geomorphologists can readily recognise the types of sources listed in Table 1. The difficulty may be in identifying whether they are active or when sediment transport last occurred. Mapping is best undertaken after a high-flow event when features and deposits are still fresh. It is suggested that field maps are produced at a large scale, combining or separately mapping the morphological features of the channel and the inputs (Fig. 5), and that these are then used to produce maps of sediment sources and of the bed and bars. Ideally, a range of characteristics can be mapped, perhaps using a series of colours/symbols to show the various characteristics. For example, for external inputs, different symbols for different types of source may be used, different colours according to sediment calibre, different sized symbols according to magnitude of input, and different density of arrows depending on likely frequency or recency of activity. Some of

these characteristics may be difficult to identify in the field and diagnostic properties would vary from area to area. The limits of the floodplain should also be mapped so that potential hillslope/channel coupling and potential external sources can be identified. As a minimum, the simple presence of such features can be mapped, indicating their potential to supply sediment.

The channel bed and bars and the floodplain are the sediment stores and their characteristics should be mapped. The main features to record are the presence of coarse sediment. Bars and bedrock are important. It is much easier to record the state of the whole channel bed in ephemeral than in large perennial channels, where it may be the calibre of emergent areas only which can be mapped without elaborate surveys and sampling. Ideally, sediment size should be systematically sampled down the channel. In practice, depending on the size of sediment available, mapping by visual categorisation of the calibre of deposits may be sufficient to distinguish the spatial variation and

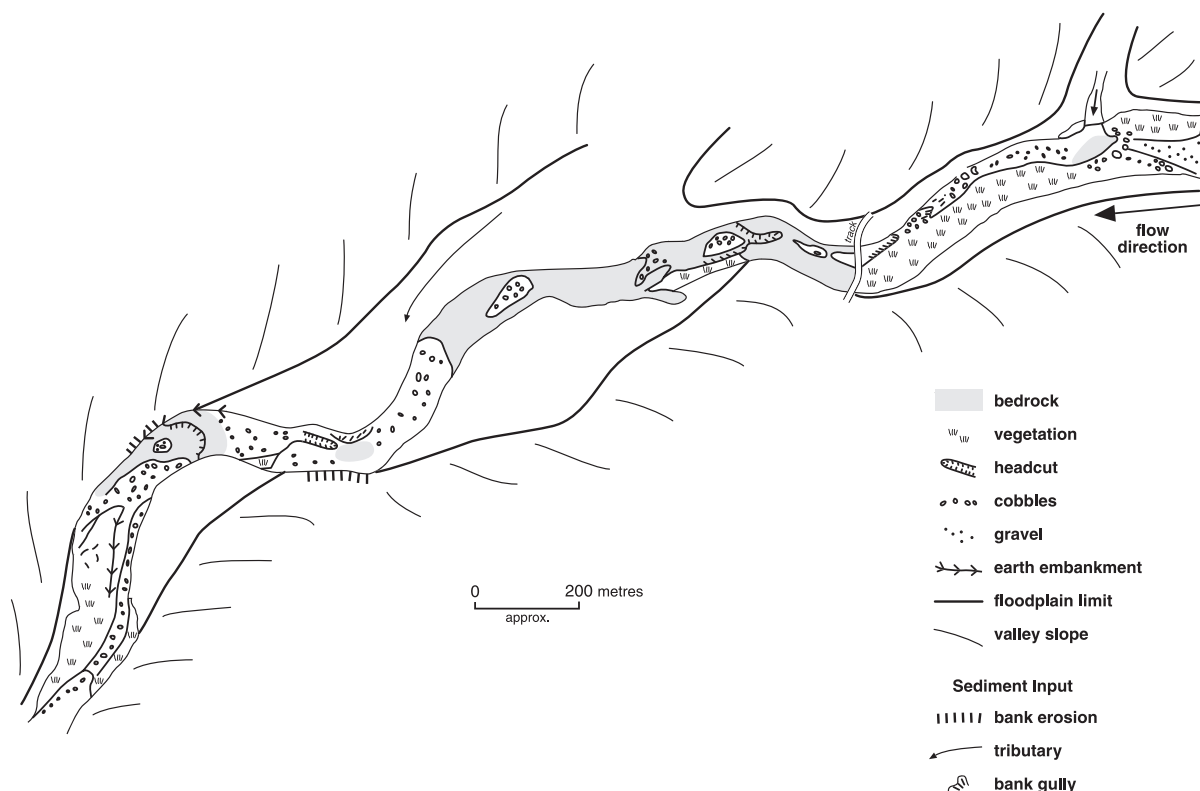


Fig. 5. Example of morphological mapping, Torrealvilla channel, SE Spain. (For location within the study reach, see Fig. 6.)

provide a framework for more detailed analysis. This method and type of analysis is most applicable and most useful in channels with a wide range of sediment size, for example a range of cobbles to silt, as is common in many ‘gravel-bed’ rivers.

The maps of sources and of stores or bed state can be combined with the morphological evidence to categorise process functioning of each part of the channel (Fig. 6). The hypothetical models outlined above can be used to make inferences on the connectivity. For example, if there is no external source within a reach but there is coarse material present, then there must be transfer within the channel from upstream. If the distance between stores is such that direct transfer is unlikely and there is little coarse material on the bed, then the degree of connectivity is likely to be low. The status of reaches can then be used to infer where breaks in transfer occur and so the connectivity of a system or a series of reaches can be mapped (Fig. 6). Limits of reaches are drawn where connectivity appears to be

broken or is doubtful. Where a linkage between reaches is not obvious, then this can be the basis for further targeted investigation (by field measurement, e.g. tracers, or hydraulic calculation or modelling) to test if sediment is being transferred.

The field mapping is undertaken for one particular time or state of the channel and the connectivity is inferred for a particular size category. Similar system connectivity diagrams could be drawn for other size fractions in the channel. Repeated mapping could then be undertaken after other events to establish the dynamics of change in connectivity or the episodicity of inputs and of coarse fraction transfers. Comparison between maps from different times would show whether connectivity remains stable, is dynamic or is propagated.

The competence of reaches can be calculated and used to check the provisional interpretation of connectivity either from peak discharge data, if available, or estimates derived from measurements of gradient, channel dimensions and estimates of roughness. Vari-

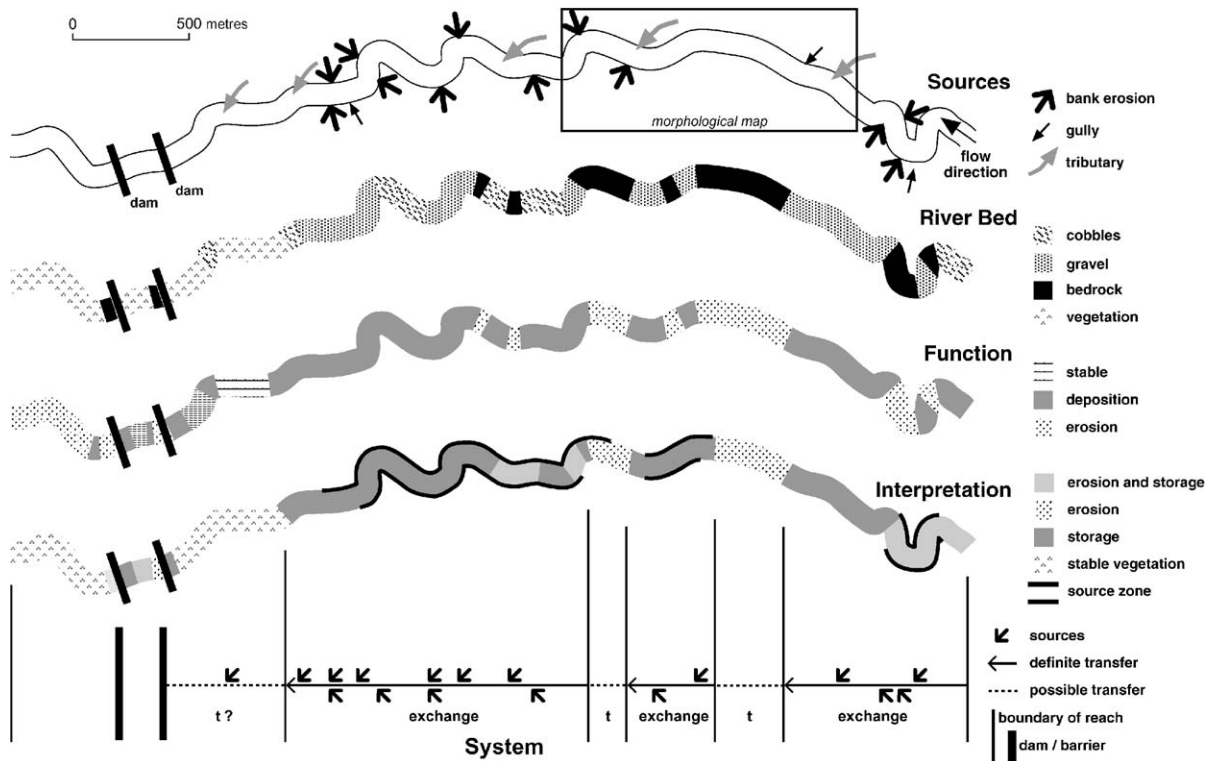


Fig. 6. Mapping and interpretation of status of reaches, Torrealvilla channel, Guadalentin basin, near Lorca, SE Spain. (Width exaggerated for representation.)

ous criteria for competence are available in the literature (e.g. Graf, 1971) and these produce a range of estimates; however, most estimates using shear stress or dimensionless shear stress show reasonable consistency. Such calculations would help to verify whether a reach of apparent lack of transfer is a non-competence reach or a flushing reach. Ancillary flood-discharge data would help to assess the time scale of connectivity for particular fractions. Long-term evidence of channel mobility and floodplain reworking, for example from historical or sedimentary evidence, would help to construct temporal scenarios of connectivity.

The methodology for mapping, compilation and analysis of evidence such that connectivity can be identified uses fairly simple techniques and easily collected data. The approach taken is very much one of a top–down or inverse approach. It examines outcomes in terms of channel morphology and sedimentology. It assumes that certain channel processes and functions can be inferred from morphological and sedimentological evidence. It is therefore truly geomorphological. Such methods are needed particularly if such analyses are to have practical applicability. It is seldom possible to instrument channels to collect actual sediment transport data, especially with sufficient density of sites to detect spatial variations. Such measurement poses special practical problems for coarse sediment load.

4. Exemplification

The system of mapping and interpretation to identify the state of connectivity in a river channel is illustrated through two examples, one an ephemeral channel in SE Spain and the other a perennial channel from NW England. Both are of moderate size (150–250 km² catchment area).

4.1. Torrealvilla

The Torrealvilla is a subcatchment of the Guadalentin in semiarid SE Spain. It has been the subject of more than 4 years of monitoring and of intensive study of the effects of flow events (Hooke and Mant, 2000). A length of several kilometres of channel has been mapped in detail. A portion of the field-scale mapping is shown in Fig. 5. These field maps have

then been used to compile the channel length maps shown in Fig. 6. A variety of channel and hillslope sediment sources are present (Fig. 6). At the time of morphological mapping, the channel varied considerably in characteristics, from bare bedrock in places to extensive coarse deposits and to vegetated sections. In the upstream part, sediment supply is relatively high but evidence of channel change, the characteristics of erosion and deposition, and measurements of size of sediment deposited, indicate that even moderate flow events are highly competent to transport coarse sediment. The evidence is that the vegetation-free parts of the channel experience high-sediment fluxes. The morphology would allow sediment storage within the channel. The bedrock reaches and those lacking bars and deposits are therefore thought to be due to sediment exhaustion. In the vegetated parts, near the downstream end of the mapped reach, transport competence would appear to be much lower. Supply is also lower since flows cannot readily erode the bed. Downstream of the vegetated reaches are check dams that have filled with material and obviously provide a barrier to coarse sediment transport under most circumstances. Overall, the status of reaches would appear to be highly variable over time but it is likely that sediment waves move through parts of the system as sediment becomes available. It is concluded that the Torrealvilla is potentially highly connected but at the time of assessment was lacking coarse sediment to produce continuous flux through the reaches.

4.2. Dane

The River Dane in Cheshire, NW England, is a very active meandering channel in a piedmont zone. Within a study length of over 10 km of channel, the sediment load ranges from cobbles to sand, the coarsest material being about 15 cm in diameter and particles of 5–10 cm diameter make up much of the bed and form many bars. The channel has been field mapped (Fig. 7) and these maps are used to compile the maps shown in Fig. 8. Almost all the coarse sediment appears to be derived from channel bank erosion; few other sources are present, except upstream channel input to a reach. There is a very close relationship between erosion and deposition sites (Fig. 8). There are also distinctive zones with no bars, which have remained relatively stable in planform over the period of historical evi-

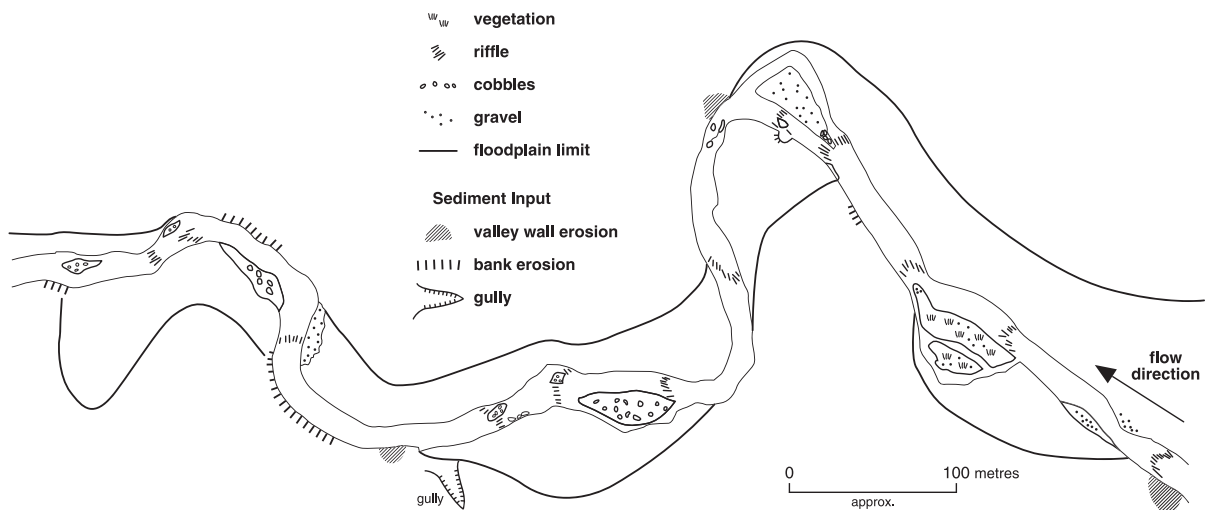


Fig. 7. Example of morphological mapping, River Dane, NW England. (For location within the study reach, see Fig. 8.)

dence (Hooke and Harvey, 1983). Questions arise over the degree of connectivity through these stable, apparently inactive reaches between other very active reaches. Pools and riffles, measured by a long-profile survey within the channel, are most abundant and well developed in the most active reaches, i.e. those with bars and bank erosion. In one reach in particular, it is doubtful whether there is coarse sediment connectivity, i.e. transfer of large cobbles. This is indicated at 'A' in Fig. 8 and is rather longer than other inactive reaches. It has no bars, has been nearly stable in position historically, and has a near uniform bed of relatively fine material, with a lack of riffles compared with other parts of the channel. It is a low-gradient reach (gradient of 0.0005) and the competence has been calculated using a range of formulae. Table 3 gives the derived values of sediment size for bankfull conditions and for a higher stage level (estimated recurrence interval ca. 5 years). The results indicate that the coarsest material would not be transported through the reach even in large flow events (Table 3). The hydraulic conditions are contrasted with typical channel dimensions and gradient for a migrating channel reach and a highly mobile reach (Table 3). It can be seen that stream power and unit stream power are also much higher for such reaches. If there were high connectivity within the channel into the upstream end of the low-gradient reach followed by a lack of transport capacity through the reach, it would be expected that signs of accumu-

lation of coarse sediment would be evident. Whilst there has been some limited increase in bend development through time (i.e. very limited propagation of bends into the upstream part of this reach), no massive stores of sediment are apparent. Furthermore, the distance of transport necessary to flush coarse material through this reach without storage is greater than the lengths of particle transport provided by tracer experiments on similar rivers (e.g. Hassan et al., 1991). Immediately downstream of the stable reach, the channel is very active but deposits of coarse material are not excessive although bank erosion does provide some material locally. Sediment deficit may be made up by a greater proportion of fine material. It is provisionally inferred from the geomorphological evidence, verified by competence calculations, that the coarsest fraction of material available in the system is not being transported through the low-gradient reach. In most other parts of this river, areas of deposition are close to zones of erosion and only short stable reaches exist. It could be that material is being flushed through those shorter reaches. A range of possibilities for coarse sediment transfer are outlined in Fig. 4. It seems possible that the transfer of coarse material in this river is very localised. Lack of propagation of change into reaches, over a 20-year period of field surveys, implies a lack of connectivity. The high lateral mobility in the upstream reach implies that the sediment goes into storage as the bars become incorporated into the flood-

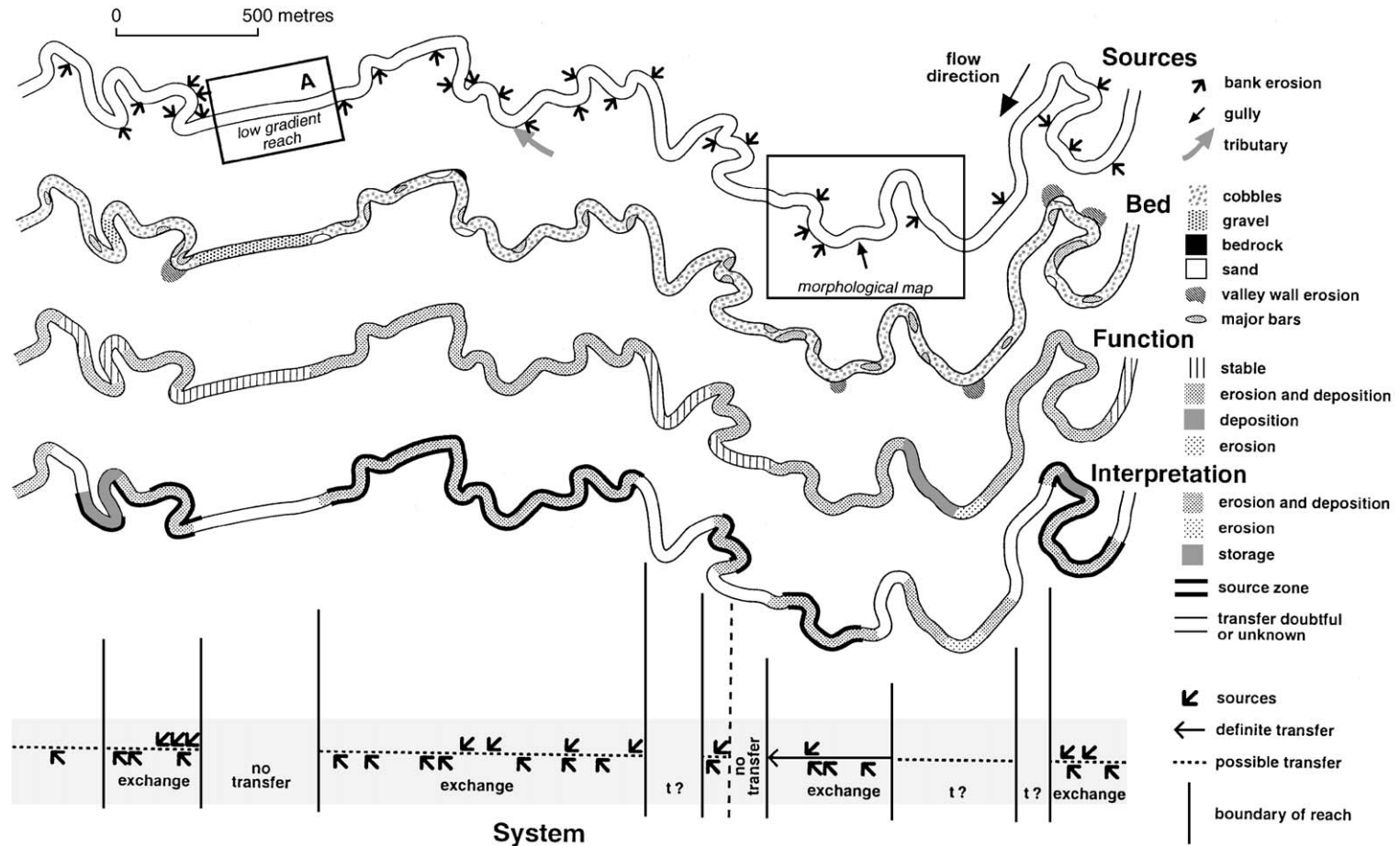


Fig. 8. Status and connectivity of reaches on the River Dane, near Congleton, Cheshire, NW England.

Table 3
Competence of reaches on the River Dane

Maximum depth (m)	Width (m)	XS area (m ²)	Perimeter (m)	Radius (m)	Slope	Manning	Velocity (m s ⁻¹)	Channel discharge (m ³ s ⁻¹)	Shear stress (kg m ⁻²)	Stream power (W m ⁻³)	Unit stream power (W m ⁻²)	Competence estimates based on:		
												Hjulstrom (1935) curves (d mm)	Critical shear stress calculation (Graf, 1971) (d mm)	Baker and Ritter (1975) equation (Max d mm)
Low-gradient reach														
(a) 2	15	30	19	1.579	0.0005	0.04	0.76	22.77	0.79	111.60	7.44	4	9	57
(b) 3	20	60	26	2.308	0.0005	0.04	0.98	58.74	1.15	287.81	14.39	8	12	70
Migrating bend reach														
(a) 1.5	20	30	23	1.304	0.0011	0.04	0.99	29.72	1.43	320.40	16.02	8	18	79
(b) 2	20	40	24	1.667	0.0011	0.04	1.17	46.70	1.83	503.45	25.17	10	20	90
Highly mobile reach														
(a) 1.5	20	30	23	1.304	0.0018	0.04	1.27	38.02	2.35	670.67	33.53	12	30	103
(b) 2	20	40	24	1.667	0.0018	0.04	1.49	59.74	3.00	1053.84	52.69	18	37	118
—bankfull Q (RI ≈ 0.3 years); b—high flow event (RI ≈ 5 years).														

a—bankfull Q (RI ≈ 0.3 years); b—high flow event (RI ≈ 5 years).

plain, but the coarse sediment is exchanged with new sources from bank erosion in the same reach. In some locations, older terraces are being eroded and these represent a much longer time scale for turnover of sediment. Under present conditions, the Dane is therefore interpreted as a series of very short unconnected or partially connected local systems in which the morphological behaviour is discrete. The reasons for the lack of connection are incompetence of transport and/or lack of sediment supply to sustain continuity.

5. Discussion and conceptual interpretation

It is suggested that the two examples exhibit differing degrees of connectivity. The River Dane has reaches in which erosion and deposition and lateral movement are very active and in which considerable amounts of coarse sediment transport take place. However, these reaches contrast with adjacent ones of long-term stability that lack indicators of coarse sediment transport. Under present conditions of morphology, sediment supply and discharge, the channel is operating as a series of discrete reaches with behaviour and coarse sediment budgets that are virtually independent of each other. On the Torrealvilla, coarse sediment appears to be carried much longer distances by moderate flood events such that all the available sediment has been removed from some reaches by a recent event. Other parts of the same channel show a lack of coarse connectivity because vegetation in the channel bed impedes erosion and movement of coarse particles. Barriers in the form of check dams are also present, which create discontinuities in the channel system. The upper part of the study length therefore exhibits potential connectivity but a present lack of connectivity because of sediment exhaustion. The lower part has been disconnected by barriers and associated immobilisation of sediment.

It is suggested that the categorisation of connectivity can provide a framework for analysis. The method provides a means of collating evidence and focuses attention on the spatial relations of the channel sediment system. The results can at least expose a lack of understanding and allow further investigation to be targeted. The method is applicable to contemporary systems. If the mapping is repeated over time, then the dynamism of connectivity can be examined. Further

validation and testing of the concepts and their application require measurements of sediment size through reaches, and ideally tracing of actual movement.

Obviously, linkages must alter in the longer term. Examination of channel connectivity and of its changes over time would allow the mechanisms and rates of propagation of change to be analysed. If a reach is presently incompetent to transport the available load, then that reach will remain as a barrier to the transmission of coarse material and propagation of change will not take place until the threshold for movement is overcome. The threshold is generally considered, according to fluvial geomorphology theory, to be overcome by deposition leading to an increase in gradient, or by channel geometry and roughness changes until unit stream power is sufficient for transport (Knighton, 1984). Reaches with hydraulic conditions exceeding the threshold for coarse sediment transport will transmit change. Those close to the threshold for transport are more likely to have a shorter lag before connectivity is made than those with conditions well below those necessary for coarse transport. Thus, the degree of connectivity in a channel has implications for responses and sensitivity to change. A highly connected channel will tend to transmit change and allow it to be propagated, whereas in an unconnected channel, change may be stalled at the limits of the connected reach and will not be propagated. Temporary storage will take place in a partially connected channel. This relates to the formation of sedimentation zones (Church and Jones, 1982) and may help in understanding the passage of sediment waves from specific inputs or disturbances, as for example in the work of Madej and Ozaki (1996) and Wathen and Hoey (1998). Harvey (2002) has explored and exemplified the effect of different types of coupling on mechanisms and time scales of propagation of change. He discusses the effects of discontinuities of sediment transport on the formation of sedimentation zones in valley floors on time scales ranging from events to decades–centuries for major influxes of sediment and much longer time scales (up to hundreds of thousands of years) for basin-scale adjustments to base-level change.

The importance of morphological variation and controls on propagation of change is being increasingly recognised, for example in the work of Fryirs and Brierley (2001), in which they identify the influ-

ence of various morphological types of channel which they term ‘River Styles’. The recognition of the lack of coarse sediment transport through some reaches is compatible with results of research on gravel–sand transitions and discontinuities in which it has been shown that these arise from a lack of coarse sediment transport particularly where there is a decrease in gradient (Ferguson et al., 1998). Other work identifies mechanisms of selective transport and the role of patches of bed material (e.g. Klingeman et al., 1998).

The system of mapping and identification of the status of reaches in the way indicated means that the spatial scale of functioning of the channel sediment system will be made explicit. Schumm (1977) identified three functional ones, source zone, transfer zone and depositional zone, and Newson (1992) recognised that fluvial systems may actually operate as a series of ‘little Schumms’; in other words, there may be a succession of erosional, transfer and depositional zones through a system. From the evidence presented here, on the River Dane, for example, these are very short sections. On other rivers, they may be much longer, related to major morphological discontinuities such as gorges. This operation of interconnected channel subsystems is already well recognised in discontinuous ephemeral channels (Bull, 1997). Thus, this approach emphasises the spatial variability of sediment transfer down river channels, and indicates that a particular flux may be very localised.

There are important implications for instrumentation and for the interpretation of measurements of bedload flux. Such measurements are valuable for validating the fluxes that occur and for understanding their mechanisms and dynamics, but care must be taken on generalising coarse sediment flux through a whole drainage system or even to other reaches from measurements taken at a single cross section. The kind of mapping and analysis suggested here can set the context for specific site measurements.

It appears that the coarse sediment flux may vary spatially, and the variations are closely related to channel morphology. Negative feedbacks are generally assumed such that the channel morphology adjusts until the channel is able to carry the sediment delivered to it. However, positive feedback can also occur such that in a highly competent reach, there is a lack of deposition so the reach remains competent and coarse bars do not form (Hooke, 1996). Likewise, if a reach

becomes vegetated or stabilised, then there may be lack of sediment supply that will be perpetuated until fresh supplies are input from upstream or external sources, or until a large event destabilises the reach. However, this becomes less likely as the sediment becomes more stabilised. Likewise, zones of deposition and bars can persist as shallowing creates hydraulic conditions for deposition.

6. Conclusions

Fluvial systems vary in their degree of connectivity at any one time. It is important to recognise that not all fluvial systems have high connectivity for coarse sediment.

A classification of types of connectivity is suggested, comprising unconnected, partially connected, potentially connected, connected and disconnected. This categorisation can provide a conceptual framework for identifying the status of river reaches and linkages between reaches and thus a basis for understanding how a reach is functioning.

Such identification makes the links and spatial relations explicit and emphasises the effects of morphology and spatial variability on sediment transfer.

Questions arise particularly on the transmission of coarse material through reaches that lack bars and locations for storage of sediment in transit and have finer material than upstream. Evidence presented here suggests that these zones may arise for several reasons:

Lack of sediment sources and supply, leading to exhaustion—producing a potentially connected system.

Flushing through of sediment in high flows—producing a connected system.

Lack of competence to transport coarse material—producing an unconnected system with localised responses and budgets.

The specific reason will differ from one reach to another and one system to another, but morphological and sedimentological evidence and the locations of sources and stores will help to identify the applicable situation.

A method of mapping and identification of functional zones and discontinuities has been proposed.

This method requires geomorphological expertise and is based on the premise that the morphology reflects the processes. Such an approach is valuable for the targeting of further research and could be useful for channel management. It provides a context for the interpretation of measurements of sediment flux.

The identification of the connectivity within a channel system has implications for understanding the propagation of changes, the sensitivity of response to perturbations and the longer-term evolution of the fluvial system.

Acknowledgements

An oral version of this paper was presented at the 5th International Geomorphology Conference, Tokyo, August 2001. I am grateful to Mike Church, John Lewin and an anonymous referee, and to the Editor, Adrian Harvey, for their constructive comments on the manuscript.

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